

DEREK TAYLOR

Geospatial Software Engineer

dkttaylor916@gmail.com • +1 (916) 214 3505 • New York City, New York • [Portfolio](#)

Software engineer with 8+ years building geospatial systems for planners and researchers, from transit routing and large-scale spatial processing pipelines to interactive planning tools in production. Has led full-stack development of web applications for transport planners and a national rail operator, and architected a Python platform that computes built environment metrics across 70+ global cities. Trained in city planning at UC Berkeley, so equally comfortable with GTFS, routing engines, and performance tuning as with the planners who use the output. Works daily with AI coding tools to move faster without lowering code quality.

WORK EXPERIENCE

Gensler Research Institute

June 2025 – Present

Lead GIS Engineer • New York City, New York

- Architected and maintain a production Python package (GeoPandas, DuckDB, H3) that computes 40 built environment metrics across 70+ global cities; cut processing from weeks to hours through spatial indexing, columnar processing, and parallelization
- Deployed OpenTripPlanner across 70+ cities to generate transit isochrones at scale, building the GTFS and OpenStreetMap ingestion pipeline and batch routing jobs that feed accessibility analysis
- Lead engineer and project manager for the Urban Intelligence Platform, a React and Mapbox GL JS web app giving researchers and design leaders direct access to city and building level analysis, census data, and location intelligence
- Built a vector tile pipeline (Tippecanoe, PMTiles) serving 30+ GB of Overture Maps and parcel data to the browser with fast, low cost map rendering
- Built and validated statistical models linking urban form to resident satisfaction, turning results into decision ready findings for design teams
- Set code standards, review, and testing practices for the team; mentored a cohort of interns through documented, repeatable data workflows

Spatial Design Hub

Aug. 2023 – June 2025

GIS Developer • London, UK

- Led full stack development for a dashboard visualizing complex rail modeling results for Network Rail UK
- Led full stack development for a web application, UrbanAccess, assisting transport planners with data analysis and site planning
- Designed and maintained data pipelines and APIs in Python (pandas) and MongoDB, improving reliability and turnaround for client deliverables
- Built interactive maps in Svelte and Mapbox GL JS with custom interaction and animation

The Pudding

Aug. 2023

Freelance Data Journalist • London, UK

- Selected as a freelance journalist to code, write, and produce [Climate Zones](#), a visual data story; storyboarded and workshopped collaboratively with Pudding staff, designed in Figma and coded in Svelte

Beyond Words Studio

Oct. 2022 – Aug. 2023

Freelance Data Visualization & GIS Specialist • London, UK

- Collected, cleaned, and analyzed data to design physical map visualizations, including a [Covid map](#) presented at an installation for the Bill & Melinda Gates Foundation
- Built a 3D elevation model of [Africa](#) (Rasterio, Blender, QGIS) linking climate, elevation, and farming systems

Goudappel

June 2018 – June 2022

Data Scientist and GIS Specialist / US Policy Researcher • Amsterdam, Netherlands

- Collected, cleaned, analyzed, and visualized real-world data from Dutch and international municipalities (including Amsterdam, San Francisco, and Gothenburg) using Python, SQL, and GIS to present social, transportation, and environmental data

EDUCATION

University of California, Berkeley

Dec. 2017

B.S., City Planning – Geospatial Information Systems and Technology • Berkeley, California

SKILLS, LANGUAGES & INTERESTS

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

Geospatial & Routing: GeoPandas, DuckDB, PostGIS, H3, OpenTripPlanner, GTFS, OpenStreetMap, Overture Maps, QGIS, ESRI suite

Web & Visualization: React, Svelte, D3.js, Mapbox GL JS, MapLibre GL JS, Tippecanoe, PMTiles, Figma

Data & Tools: PostgreSQL, MongoDB, pandas, Git/GitHub, AI coding tools (Claude, Copilot)

Interests: Biking, surfing, playing drums, saunas, chess, jumping into cold bodies of water